

Computer Network, Magh 29th - Fall 2024

Total marks: 100, Pass marks: 45

BSE V semester

Attempt all the questions.

1. a) What is computer network? Distinguish between OSI and TCP/P reference model. [2+5]
b) What is ISDN? Explain about the ISDN architecture in detail with example. [2+6]

2. a) What is transmission media? Explain about any two-transmission media in detail. [2+5]
b) What are the major functions of data link layer? Briefly explain CSMA/CD with a suitable diagram [3+5]

3. a) Discuss about the significance of routing in computer networking? Differentiate between link state routing and distance vector routing. [2+5]

b) Explain the terms. (4x2)

i) Simple Mail Transfer Protocol ii) Domain Name System

4. a) A 7-bit Hamming Code is received as 1011011. Assume even parity and state whether the received code is correct or wrong? If wrong locate the bit in error. [7]
b) What is congestion control? Explain about leaky bucket algorithm of traffic shaping. (2+6)

OR

Briefly explain the various TCP congestion control policies.

5. a) Why port number is used in networking? Distinguish between TCP and UDP. How is TCP connection established? Explain. [2+2+4]

OR

Define DHCP, SMTP, POP and IMAP with their port number.

- b) What are the drawbacks in IPV4? Which of these drawbacks do IPV6 solve? Explain [2+5]

6. a) What is network security? How can different types of firewalls enhance network security? (1+2+5)

- b) What is HTTP protocol? With an example explain how a request initiated by a HTTP client is served by a HTTP server [2+5]

7. Write short notes on (any two) (2*5)

- a) Recent Trends in Telecom Technologies: 2G/3G/4G/5G.
b) Switch
c) NGN
d) NAT

a -> Network
b -> Switch
c -> Internet
d -> Link
e -> Sender